

LIU FENG

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Self Introduction

I am passionate about technological progress and have an unrelenting pursuit of knowledge, particularly in computer science. I enjoy meeting interesting people and exploring diverse facets of society, staying curious about cultures and phenomena across the world. Outside of work, I am an avid soccer fan (Real Madrid) and enjoy hiking and skiing. I previously worked on full-stack development with Node.js, but with the rise of AI, I have been self-learning LLM-related technologies and plan to transition my career toward AI.

Education

Anhui University (211) the gifted class of computer science

Hefei, China

2017.9 - 2021.6

- GPA:3.30/4
- Languages: Chinese Grade A, English IELTS 6
- Programming Capabilities:CCF-CSP 180
- Relevant Courses:Principles of Computer Organization,Data Structures,Operating Systems,Computer Networks,Database Principles,Object-Oriented Programming,Digital Image Processing,Big Data Technology,Machine Learning,Pattern Recognition,Optimization Methods,Mathematical Modeling (Advanced Class),Artificial Intelligence,Principles of Compilers,Writing International Academic Papers (Graduate Course)
- Honors and Awards:Second-Class Scholarship for Academic Excellence,Anhui Province Big Data Competition

Hongkong University (QS 11) Computer Science and technology

Hongkong SAR

2025.9 - 2026.9

- Course: «High-performance computer architecture» , «Neural networks, fuzzy systems and genetic algorithms» , «Computer programming for product development and applications» ,COMP7404 «Computational intelligence and machine learning»
- Dissertation: Large language model based multi-agent materials scientist for advanced materials design

Technical Skills

• Programming:

- Proficient in Python, TypeScript, JavaScript; experience with C++, Java, Matlab, HTML, LaTeX, Typst, Rust.
- Skilled in React, UmiJS, Next.js, and SSG/SSR/CSR rendering.
- Experienced with ECharts, Ant Design, TinyMCE customization, and low-code platforms.
- Knowledge of Linux, Nginx, Docker, Git, CI/CD.
- Familiar with Taro, React Native, Weex; basic SpringBoot, MyBatis, RESTful API development.

• Tools:

- Knowledgeable in Transformer, Diffusion, VAE, DDPM, LoRA, KV Cache; familiar with LangChain, Hugging Face, LlamaFactory.
- Proficient in front-end engineering, Serverless, Sass, Babel, Webpack, Vite, NPM, Grunt.
- Experienced in Alibaba Clam system: Def API, Ali Node, Sunfire, EagleEye, end-to-end troubleshooting.
- Skilled in Node.js / BFF development, monitoring, performance tuning; proficient in MidwayJS, NestJS; understand browser rendering & V8 engine.

- Experienced in MySQL, SQL Server, TypeORM; familiar with RestAPI, GraphQL, gRPC.
- Hands-on LLM experience: fine-tuning, LoRA, KV Cache, attention mechanisms; built RAG & data analysis agents with LangChain.

Project Experience

Bybit

2025.10 - 2026.5

Developer(Intern)

- Developed Bybit's official homepage with Next.js 14, supporting 60+ languages, regional compliance rendering, and real-time market data, achieving LCP < 2.5s for millions of daily PV.
- Built a 28-app monorepo (650k+ LOC) and 14 shared libraries for Bybit's growth platform; delivered dynamic page builder, 30-language i18n, and data tracking system, boosting efficiency by 40%.

Alibaba. Zhejiang Fliggy Network Technology Co., Ltd.

2023.9 - 2024.5

Frontend Developer and Operations (Ops)

- LLM Fundamentals
 - Familiar with Transformer and Diffusion technologies; experienced in model fine-tuning, LoRA, KV cache, VAE, and DDPM; knowledgeable about various attention mechanisms, as well as Hugging Face and LLaMA-Factory.
 - Applied LangChain for RAG integration and developed data analysis agents with LangChain.
- Development of the Backend System “Jellyfish” for Risk Control
 - Included API calls for data and permission interfaces, and integration of event tracking.
- Independently developed a policy document parsing tool.
 - Automated the parsing of airline policies from Excel, achieving a coverage of 10 airlines by February with an automation rate of 75%.
 - Supported policy document types include Word 2007, Word 2003, Excel 2007, and Excel 2003.
 - By building automatic parsing capabilities for airline policy documents, improved operational efficiency, reducing the time required to process complex policy documents from one day to within two hours.
- Developed basic pages for the backend ticketing operations system.

China Merchants Bank. CMB Technology (Shenzhen) Co., Ltd.

2021.7 - 2023.9

Developer

- After joining the company through campus recruitment, I participated in the development of the cloud operations platform, specifically the OracleDM platform (a platform for operating Oracle databases). I primarily served as a frontend developer, using React 17, TypeScript, and Ant Design for web development and API data acquisition.

Research Experience

Vision-Language Model Fine-Tuning and Reinforcement Learning for Materials Characterization

Hong Kong 

2025.10-2026.05

This project forms the core research of my master's dissertation «Large Language Model Based Multi-Agent Materials Scientist for Advanced Materials Design» .

- Built a complete materials science VLM training pipeline based on Qwen2.5-VL-7B-Instruct, completing SFT + RLHF two-stage training on a single RTX 4090 (24GB)
- Constructed a dataset of 25,263 high-resolution material microscopy images (sourced from OmniScience), using LLaMA-Factory for LoRA fine-tuning (rank=8, BF16, FlashAttention-2, paged AdamW 8-bit); training loss reduced to 0.1762
- Designed a custom reward function with 6 physics-aware dimensions (crystal plane index matching, defect terminology coverage, physical law consistency, etc.) to address catastrophic forgetting during SFT; completed RL using GRPO algorithm via ms-swift
- Post-RLHF: defect recognition accuracy reached 94.3%, chart analysis capability recovered to 81.9%, physical law violation rate reduced to 8%

- All code, model weights, and datasets are open-sourced: [GitHub](#) · [HuggingFace Models](#)

Multi-Objective Complex Attribute Network Community Detection

Published a paper during undergraduate studies: «A Reduced Mixed Representation Based Multi-Objective Evolutionary Algorithm for Large-Scale Overlapping Community Detection» (DOI: 10.1109/CEC45853.2021.9504894) . This work was conducted at the Key Laboratory of Computational Intelligence and Information Processing, Ministry of Education, Anhui University.

Hefei 🇨🇳

2019-2021